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Interactive Prisoner's Dilemma using Game Theory and Decision Analysis Techniques

Abstract

This project presents an interactive simulation of the Prisoner's Dilemma, a fundamental problem in game theory. The model is extended into a multi-round format with real-life scenario-based decisions. The system supports Human vs Human and Human vs Computer modes. Game theory concepts such as Nash Equilibrium, Dominance Analysis, and Mixed Strategy Optimization are used. Gambit software is utilized to analyze equilibrium, while Python is used for simulation. The project demonstrates how cooperation evolves and how rational decisions impact outcomes over repeated interactions.

Introduction

Game theory is a mathematical framework used to model strategic interactions between rational players. The Prisoner's Dilemma is one of the most studied problems in game theory. This project enhances the classical model by introducing multiple rounds and real-life scenarios.

Objectives

- To simulate Prisoner's Dilemma in an interactive environment
- To analyze decision-making strategies
- To study Nash Equilibrium and dominant strategies
- To compare human and AI behavior

Methodology

The game is modeled using a payoff matrix. Players choose between cooperate and defect options in multiple rounds. Gambit software is used to compute equilibrium, while Python is used for simulation and visualization.

Features

- Interactive scenario-based gameplay
- Human vs Human and Human vs AI modes
- Multi-round simulation
- Strategy-based AI
- Graphical output and analysis

Technical Complexity

The project involves moderate complexity, including game modeling, simulation logic, strategy implementation, and equilibrium computation using Gambit.

Conclusion

The project successfully demonstrates strategic decision-making and highlights the importance of cooperation in repeated interactions.